

Dear Editor,

We are pleased to submit “Confinement-loss prediction in photonic crystal fiber SPR sensors: kernel surrogates versus generative data augmentation” for consideration in Signal, Image and Video Processing. The manuscript is an extended version of our paper presented at SIU 2026, prepared following the invitation extended to authors of that conference.

Designing photonic crystal fiber based surface plasmon resonance sensors requires repeated full-vectorial finite element simulation, and that cost, not any modelling difficulty, is what limits practical design. The established remedy is a machine-learned surrogate; because deep surrogates need more data than simulation can cheaply supply, recent work has turned to generative adversarial networks to synthesise the missing samples.

We show that this step is unnecessary for models suited to the data regime, and that it is actively harmful. A Bayesian-optimised support vector regressor trained on 432 finite element samples alone reaches a mean squared error of 0.121 against 0.314 for the published GAN-augmented deep network. Adding the same synthetic samples raises that error to 0.197 and causes Gaussian process regression to fail outright. Rather than report the failure, we identify its mechanism: generated sample pairs that are near-identical in the input space yet carry incompatible targets, which drives the covariance matrix towards singularity. Timing measurements show the generative step to be the most expensive stage of the pipeline while reducing accuracy.

The work also yields design guidance. Shapley attribution attributes confinement loss primarily to the outer cladding hole diameters, consistent with their role in gating evanescent leakage, and a Monte Carlo analysis identifies the lattice pitch as the dominant source of resonance instability under fabrication tolerance. We believe the results will interest readers working on machine-learned surrogates for physical systems more broadly, since the augmentation failure we characterise is not specific to this sensor.

Relative to the conference version, the article adds per-geometry cross-validation of every model, the mechanistic account and supporting evidence for the Gaussian process failure, the manufacturing-tolerance analysis, measured training and inference costs, and an expanded treatment of related work and the evaluation protocol.

The manuscript is prepared in the Springer Nature template with the [iicol] option and is 10 pages, with supporting material supplied as Online Resource 1. All figures and tables have been newly prepared for this submission and none is reproduced from the conference paper. The work is original, is not under consideration elsewhere, and all authors have approved the submission.

Thank you for your consideration.

Yours sincerely,

Mohammad Khaleel Hammoud

On behalf of Cem Kalyoncu and Ahmet Yaşlı

Department of Computer Engineering, European University of Lefke

mhmdkhalil.h@gmail.com